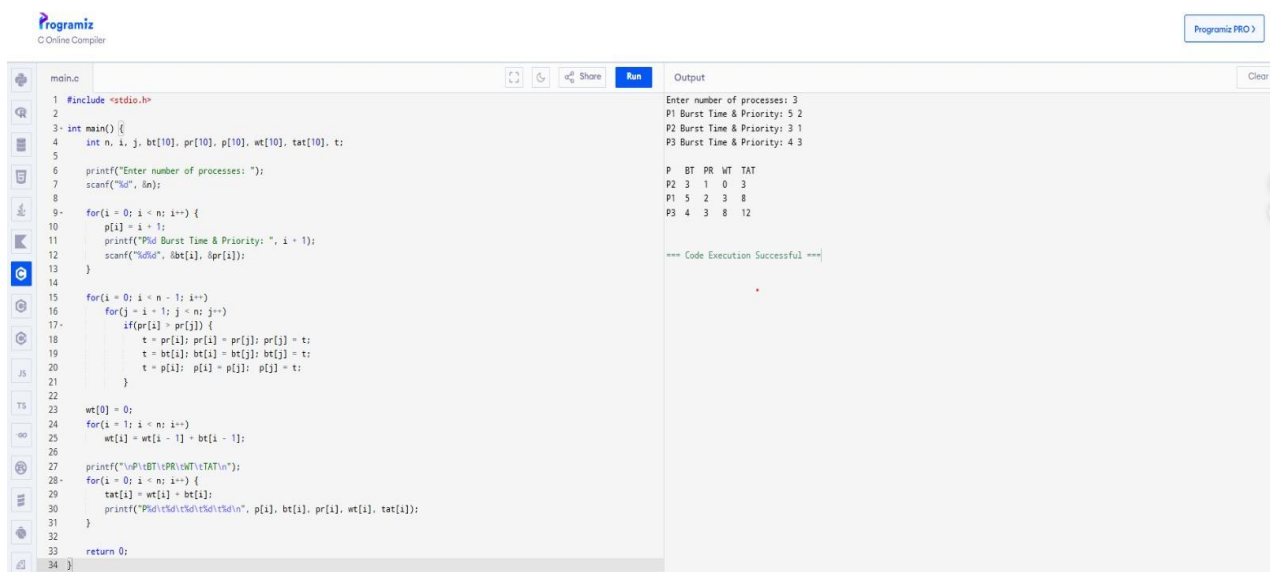


QUESTION 5:

Construct a scheduling program with C that selects the waiting process with the highest priority to execute next

CODING:



The screenshot shows a C program in a web-based IDE. The program implements a scheduling algorithm where processes are selected based on their priority. The code includes a main function that takes the number of processes (n) and their burst times (bt) and priorities (pr) as input. It then calculates the waiting time (wt) and turnaround time (tat) for each process. The output shows the execution details for three processes: P1, P2, and P3.

```
1 #include <stdio.h>
2
3 int main() {
4     int n, i, j, bt[10], pr[10], wt[10], tat[10], t;
5
6     printf("Enter number of processes: ");
7     scanf("%d", &n);
8
9     for(i = 0; i < n; i++) {
10         p[i] = i + 1;
11         printf("P%d Burst Time & Priority: ", i + 1);
12         scanf("%d%d", &bt[i], &pr[i]);
13     }
14
15     for(i = 0; i < n - 1; i++)
16         for(j = i + 1; j < n; j++)
17             if(pr[i] > pr[j]) {
18                 t = pr[i]; pr[i] = pr[j]; pr[j] = t;
19                 t = bt[i]; bt[i] = bt[j]; bt[j] = t;
20                 t = p[i]; p[i] = p[j]; p[j] = t;
21             }
22
23     wt[0] = 0;
24     for(i = 1; i < n; i++)
25         wt[i] = wt[i - 1] + bt[i - 1];
26
27     printf("\nP\tBT\tPR\tWT\tTAT\n");
28     for(i = 0; i < n; i++) {
29         tat[i] = wt[i] + bt[i];
30         printf("P%d\t%d\t%d\t%d\t%d\n", p[i], bt[i], pr[i], wt[i], tat[i]);
31     }
32
33     return 0;
34 }
```

Output:

```
Enter number of processes: 3
P1 Burst Time & Priority: 5 2
P2 Burst Time & Priority: 3 1
P3 Burst Time & Priority: 4 3

P  BT  PR  WT  TAT
P2  3  1  0  3
P1  5  2  3  8
P3  4  3  8  12

=== Code Execution Successful ===
```